

Towards Cultural Inclusion: Using Mobile Technologies to Increase Access to Audio Description

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Abstract: This paper describes a National Park Service (NPS) and University of Hawaii research project that is developing a mobile application for audio describing NPS print brochures for blind and visually impaired park users. The project has the potential to expand access to cultural and aesthetic material for blind and visually impaired people.

Key words: accessibility, assistive technology, blind and visually impaired

Introduction

Creating access for people with disabilities by public and private entities in the United States is most often driven by the compulsion to meet the minimum requirements of the Americans with Disabilities Act (Jones, et al., 2012) and other Federal and State civil rights laws. Often the focus of accessibility efforts is on the obvious and the essential. Can someone in a wheelchair get into a building to conduct business? Can someone who is blind or visually impaired independently ride the bus or read a government webpage? Access to cultural and aesthetic content is usually seen as a luxury rather than a right. This oversight deprives many people with disabilities from social inclusion, recreation, and the benefits that cultural and aesthetic pleasures bring to an individual's quality of life.

Audio description is a means of providing people who are blind and visually impaired with a verbal synopsis of visual content. Audio description has been most widely utilized as a narrative technique for providing visual access to live cultural events such as movies and theatrical performances. However, audio description is largely underutilized for more static visual material such as museum displays, outdoor attractions, and image-dependent brochures. The lack of audio-described material is due in part to the time-consuming and specialized process of describing visual content using human actors.

The National Park Service (NPS) is funding a research team from the University of Hawaii at Manoa to use mobile technology as a platform to offer audio described park service brochures for people who are blind or visually impaired. Mobile technologies make it possible to more efficiently and affordably create audio description content that is uniform, portable, and easily adjustable to meet the needs of individual users. Challenges include developing a best

practices protocol on audio description for NPS personnel, applying accessibility guidelines, and developing a user-friendly application for both the content providers and blind and visually impaired audiences.

Need for Alternate Format Brochures

The NPS has 342 brochures that are designed and produced in collaboration with their publication house in Harper's Ferry, West Virginia. The NPS uses brochures not only to convey basic information about the park, such as park hours, hazards, and the location of campsites and visitor centers, but also to give the visitor a sense of the historical, natural, and cultural significance of the park. Increasingly, this is done through the use of highly visual media such as photographs, maps and drawings. NPS brochures are seen not only as a tool for use by visitors while they are in the park, but as an aesthetically pleasing souvenir for park users to take away with them when they leave (Hartley et al, 2015).

Currently the NPS offers large print, some Braille, and a few audio described brochures for blind and visually impaired people. The Americans with Disabilities Act (ADA) and Section 508 of the Rehabilitation Act require federal agencies to make alternative and accessible formats of information offered in print and on the web (Schuur, 2001). There are no specific mandates about best practices for achieving high quality print access (Lazar, Goldstein, & Taylor, 2015). According to the American Foundation for the Blind (AFB, n.d.), in 2012 there were over 20.3 million adult Americans with a visual impairment. In keeping with the NPS effort to offer broad access to the park system to a wide range of people, the NPS is actively developing alternative formats for their Park site visitor displays, videos, and printed materials.

Project Goals and Description

With the increasing popularity of mobile devices in the United States, using software programs via mobile technologies, such as smartphones and mobile apps, offers the opportunity to engage with alternate, and potentially accessible, applications for personal and on-demand use. Bouyed by the growing use of mobile technologies, this NPS project will develop an application for delivering audio-described Park Service brochures in an economical and efficient mode for blind and visually impaired park users.

There is limited research and real-world guidelines for creating audio described print materials (Braun, 2011; Morales, 2011). One of the NPS project's goals is to develop a best practices protocol for creating usable and current output in a repeatable and consistent manner. Existing research and documentation falls into three categories (Szarkowska, 2011): 1. soundtrack audio description options for film and television production; 2. live theater

performance audio described recordings, and 3. museum tour guided programs directed toward blind and visually impaired visitors. There are currently no national or international standards for providing consistent and high quality audio description (Orero & Vilaró Soler, 2012; Morales, 2011).

The backend software program for creating content for audio output to a mobile device application will require NPS personnel to input data into an online tool, created during this process, providing an opportunity for step-by-step instructions based on a set of best practices. This NPS project will implement recognized international accessibility guidelines from the World Wide Web Consortium (W3C) Web Accessibility Initiative (WAI) (www.w3c.org/WAI) to bring the software applications in compliance with the ADA and Section 508 regulations and will be used to contextualize the final best practices documentation. However, the project also will go a step further by collecting, analyzing and integrating the existing ad hoc best practices in audio description with project research on best practices so that blind and visually impaired end-users will have a high-quality audio-described brochure that provides them with the same informational, aesthetic and cultural experience intended for sighted users.

Finally, the NPS project's software prototypes will be tested by NPS personnel and blind and visually impaired users at three NPS sites for evaluation and feedback. Using real NPS sites for testing will bring together best practices protocols for creating content with blind user preferences for access and engagement. Our hope is that such research and testing in practical situations will provide an opportunity for rich dialog on improving audio description and mobile technology and thus contribute to the knowledge base on audio description.

Implications

Providing an audio-described brochure for blind and visually impaired visitors will expand NPS reach to people with disabilities wishing to experience and enjoy the parks as other visitors currently do. The National Parks offer a social context that is supposed to be educational, historical, and entertaining for all people. This project has the potential to significantly enhance the park experience for blind and visually impaired visitors by enabling them to explore and engage in the social and cultural opportunities that the parks provide. This project also will put the NPS at the forefront of the accessibility field, as media becomes increasingly visual and less dependent on the written word for communicating to the general public.

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~~Out date: 5 Oct 2023, 09:35AM Mountain Standard Time~~

~~Due date: 11 Oct 2023, 11:59PM Mountain Standard Time~~

- ~~• This lab is worth 100 points.~~
- ~~• Please submit this document with all the questions answered along with a copy of your Orange pipeline before the due date in WyoCourses. File should be named as lab2.1_lastname.docx / lab2.1_lastname.ows.~~

Objective: ~~To build a modeling pipeline in Orange and apply multi-linear regression modeling to predict a quantitative target variable.~~

Data:

~~Horizontal drilling which provides additional exposure of the reservoir to the wellbore (lateral length) is one of the primary drivers that made economic production of oil and gas from the tight shale reservoirs in the US a success, in conjunction with stimulation using hydraulic fracturing.~~

~~Objective of the exercise is to understand effect of lateral length of a horizontal well on shale gas production in some of the US shale plays using multiple linear regression analysis. While lateral length would be one of the primary predictors, other variables considered are vertical section length of~~

the horizontal well, proppant volume and fluid volume pumped for stimulating the horizontal section of the well using hydraulic fracturing and type of shale plays.

Data source: <https://doi.org/10.15786/22394989> [WyoScholar Digital Repository] Originally downloaded from National Oil and Gas Gateway (<http://www.noggateway.org/explore>). From participating states that contribute to this data source, Arkansas and Colorado were identified as states of interest for the project owing to the level of shale gas drilling and production activity in these states over the period 2008-2018 and data availability on Fluids and Proppants.

Variables of interest:

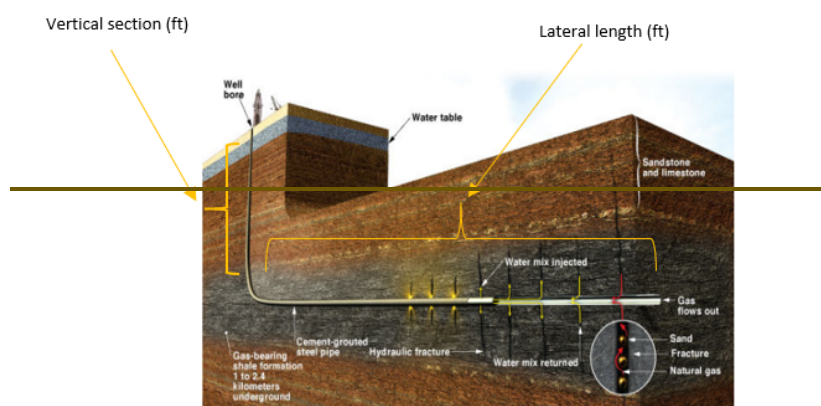


Figure 1: Horizontal well schematic (from The Academy of Medicine, 2017)

Referring to the Figure-1 above, the following were identified as variables of interest after reviewing the dataset:

Response (Target) variable: Max_Gas, Maximum Gas Production (Million Standard Cubic Feet, MSCF)

Since shale gas wells have their peak production in the first 2-3 years after the well is drilled, stimulated and put on production, maximum annual gas production is selected as the best response variable to predict using the predictor variables identified below

Predictor variables:

Lat_Len, Lateral length of the horizontal section (feet)

Measured Depth, Total Depth of the well (feet)

Bottom and Top Perforation depth (feet)

Wellhead Latitude and Longitude (decimal degrees), surface location of the well

Fluid, Total amount of fracturing fluid (typically water) pumped to create the hydraulic fractures (Barrels)

Proppant, Total amount of proppant (typically sand) pumped along with the fracturing fluid as a slurry to keep the hydraulically created fractures open (Pounds)

ShalePlay (a dummy variable, 0- Fayetteville, Arkansas Shale gas play and 1- Mancos, Colorado Shale gas play)

Drilling Orientation (a dummy variable)

Data: For this lab, please download *final.xlsx* to your local folder.

Note: This is the final processed dataset from the raw files in the data source link using Python

Lab Instructions

1. Open Orange, and add **File** widget to the canvas and load the *final.xlsx* dataset.

File (1) - Orange

Source

File: final.xlsx

URL:

Reload

File Type

Automatically detect type

Info

6553 instances
16 features (0.7% missing values)
Data has no target variable.
1 meta attribute

Columns (Double click to edit)

	Name	Type	Role	Values
1	Feature 1	numeric	feature	
2	API NUMBER	numeric	feature	
3	STATE	categorical	feature	Arkansas, Colorado
4	WELLHEAD	numeric	feature	
5	LATITUDE, ...	numeric	feature	
6	WELLHEAD LONGITUDE, ...	numeric	feature	
7	MEASURED DEPTH, FEET	numeric	feature	
8	DRILLING ORIENTATION	categorical	feature	Directional, Horizontal, Vertical
9	TOP WELL PERFORATION, ...	numeric	feature	
10	BOTTOM WELL PERFORATION, ...	numeric	feature	
11	BOTTOM HOLE LATITUDE, ...	numeric	feature	
12	BOTTOM HOLE LONGITUDE, ...	numeric	feature	
13	ELEVATION	numeric	feature	
14	Fluid	numeric	feature	
15	Proppant	numeric	feature	
16	LATLEN	numeric	feature	
17	MAXGAS	numeric	feature	
18	WELL NAME	text	meta	

2. Inspect the **File** widget and complete the following table. (10 points)

Total instances	
Dimensionality of the data set	

3. Add **Feature Statistics** and **Data Table** widgets to the **File** widget. Inspect the features using these two widgets. (10 points)

List down key observations about feature inconsistencies and possible actions to clean out the dataset

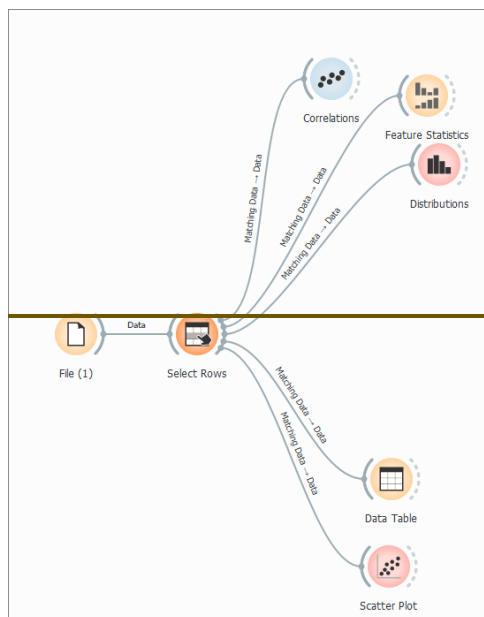
4. Add **Select Rows** widget to the **File** widget and apply the following filters:

The screenshot shows the 'Select Rows - Orange' widget window. It contains a table of conditions for filtering data rows. The conditions are as follows:

Variable	Operator	Value	Action
MEASURED DEPTH, FEET	is greater than	0.000000	X
Fluid	is greater than	0.000000	X
Proppant	is greater than	0.000000	X
MAXGAS	is greater than	0.000000	X
LATLEN	is greater than	0.000000	X
LATLEN	is below	25000	X
MEASURED DEPTH, FEET	is below	25000	X

Below the table, there are buttons: 'Add Condition', 'Add All Variables', and 'Remove All'. At the bottom, there are checkboxes for 'Remove unused features' and 'Remove unused classes', and a 'Send Automatically' button with a checked checkbox. The status bar at the bottom shows a question mark icon, a document icon, and the text '6553 | 6013 | 540 | 6553'.

5. Add **Data Table**, **Feature Statistics**, **Distribution**, **Scatter Plot & Correlations** and **Scatter Plot** widgets to complete the table below: (30 points)



Min and Max of key predictor variables

Distribution of the predictor variables:

Min, Max and distribution of the target variable, Max_Gas

% of wells from the two states (shale plays) and drilling orientation

Any rows missing target values?

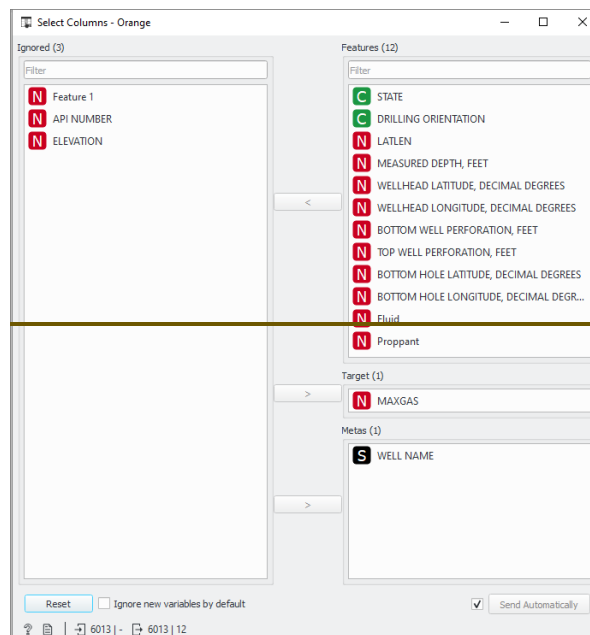
From the **Scatter Plot**, what are the predictors that you believe would have the highest impact on predicting Max_Gas?

How would you describe the relationships for these predictors with the target?

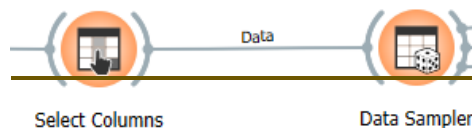
Do you notice outliers in the data?

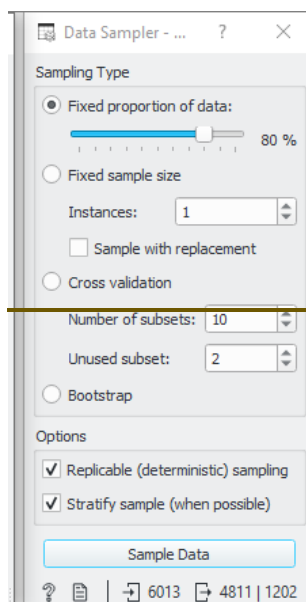
From the **correlations** widget, what do you think about the correlation between predictors?

6. Add and open **Select Columns** widget and confirm selection as below:

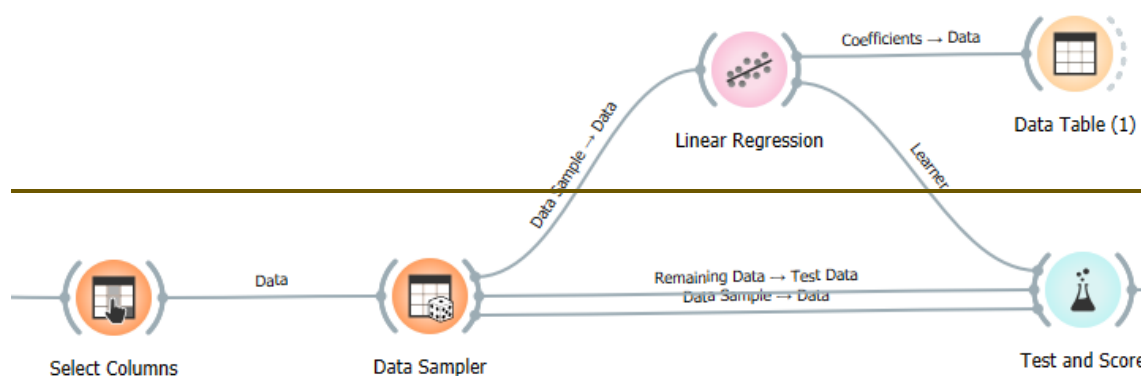


7. Add the **Data Sampler** widgets to the **Select Columns** widget as shown below:





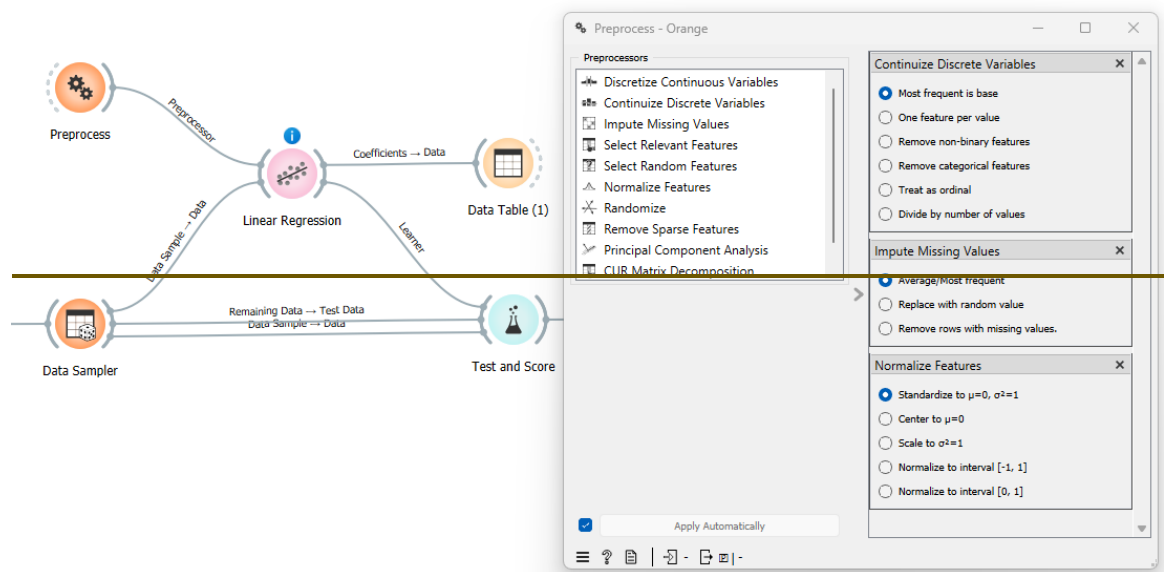
8. Add the **Linear Regression**, **Test and Score** widgets and connect the **Data Sampler** widget to **Test and Score** widget as shown below. Also, add a **Data Table** widget to the **Linear Regression** widget.



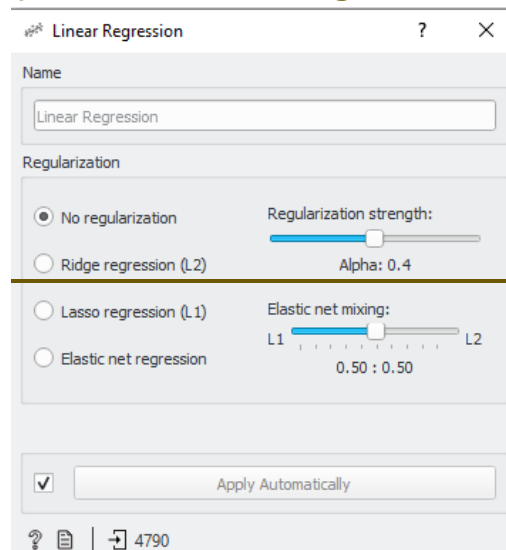
9. Default preprocessing is applied by the **Linear Regression** widget in the following order:

- Remove instances with unknown target values.
- Dummy code categorical variables (with one-hot encoding)
- Remove empty(null) columns.
- Impute missing values with mean values.

If we need to modify any of the above, we can add a **Preprocess** widget to the **Linear Regression** widget as shown below:



10. Open and inspect the **Linear Regression** widget:



11. Open **Data Table** widget connected to the **Linear Regression** widget and inspect the parameter coefficients.
(10 points)

	name	coef
1	intercept	379986
5	LATLEN	83916.2
9	BOTTOM WELL ...	70375.9
14	Proppant	14711.5
7	WELLHEAD LAT...	8573.99
13	Fluid	7569.95
10	TOP WELL PERF...	-740.1
4	DRILLING ORIE...	-43099.9
6	MEASURED DE...	-43397.8
2	STATE=Colorado	-104544
11	BOTTOM HOLE ...	-116392
3	DRILLING ORIE...	-118728
8	WELLHEAD LO...	-118756
12	BOTTOM HOLE ...	-174648

What can you say about the relationships between the different predictors and the response variable (Max_Gas)?

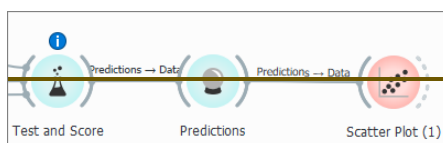
12. — Open **Test and Score** widget and complete the table below:
(15 points)

What are the evaluation metrics you see?

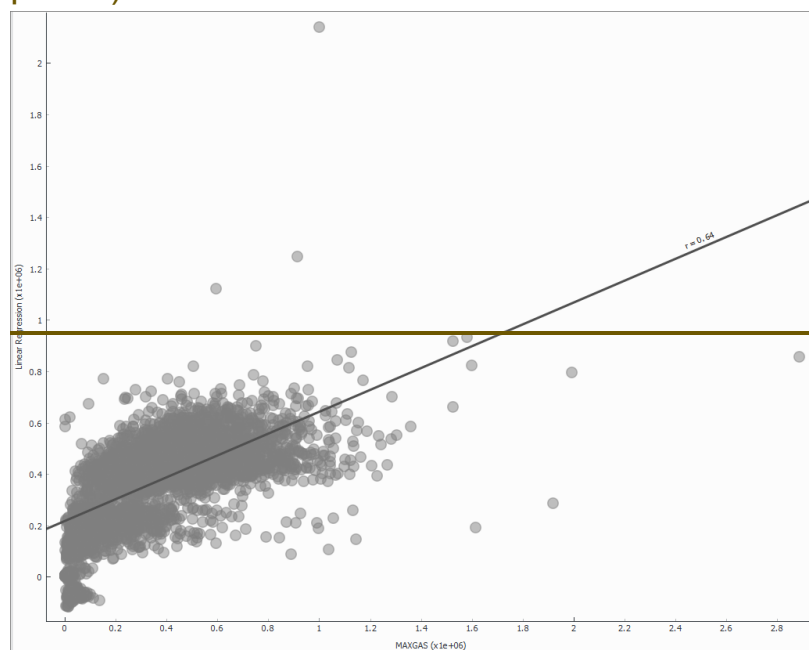
Model	RMSE	MAE	R2
Linear Regression (cross validation, 10 folds)			
Linear Regression (Test on test data)			

How would you interpret this model based on R2 and cross validation?

13. Add **Predictions** and **Scatter Plot** widgets to the **Test and Score** widget as shown below:

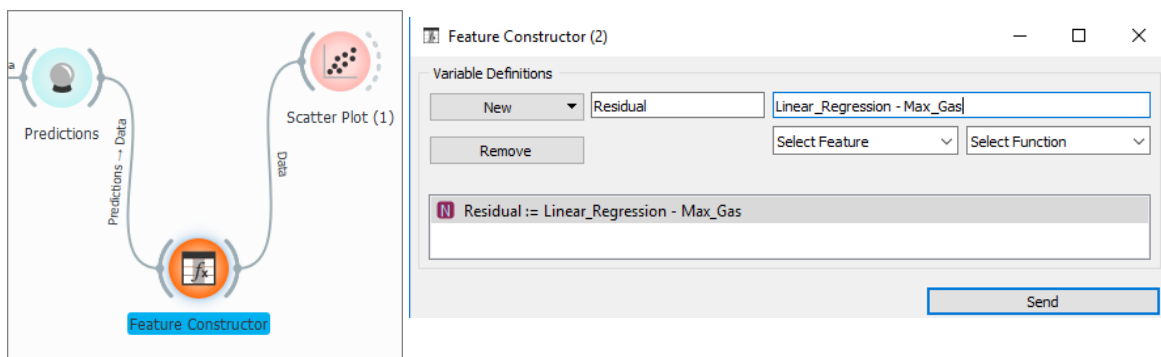


14. Open the **Scatter Plot** widget connected to **Predictions** widget. Select **Max_Gas** as X-axis and **Linear Regression** as Y-axis. (10 points)

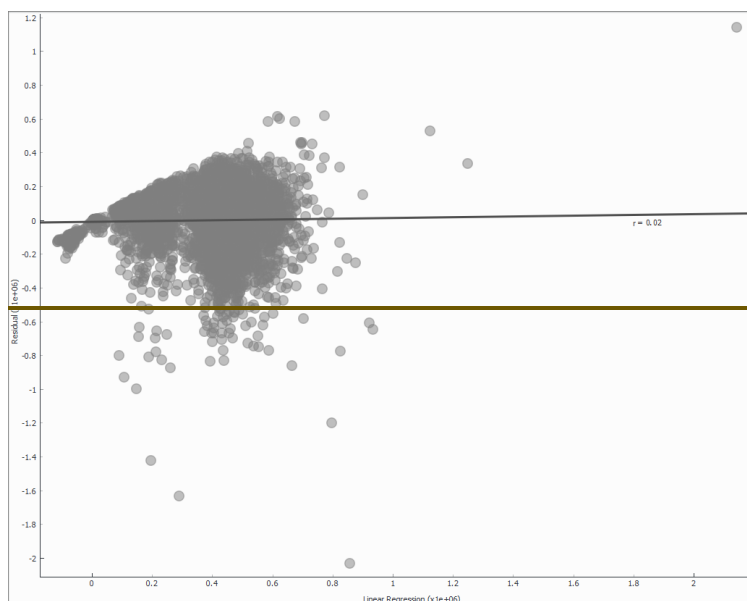


What do you think about the prediction quality?

15. As part of model diagnostics in linear regression, it is standard practice to visualize the residual (error) plot and look for patterns to understand model validity. Residual variable is constructed as shown below using the **Feature Constructor** widget added to the **Predictions** widget as shown below:



16. Open **Scatter Plot** widget and select Linear Regression (Predicted Max_Gas from the regression model) and Residual. Plot would look as below:
(15 points)



Do you see a pattern?
What does this indicate?
What can you do to address this issue?